FUNDAMENTALS OF LOGIC

Discrete Mathematical structure is a study of discrete structures which are abstract mathematical model that deals with discrete objects and their relationships.

A Logic is used as a tool in no. of situations for variety of reasons. A logic is found to be extremely useful in decision making. Logic is the science dealing with the methods of reasoning.

Applications of logic in the field of CS&E :

① To check the correctness of a program.

② Analysis of algorithms and their implementations

③ Development of algorithms into a structured program in a programming Language.

④ AI, computer networks, OS, Theory of computations, fuzzy logic & in some of the functioning of expert systems.

Basic terminologies

① Propositions : (Declarative statement, it can have)

A proposition is a declarative sentence which can be assigned the value true (T) or the false value (F). The true or the false value assigned to the proposition is called the truth value.

A proposition is said to have T if it is true and it is said to have F if it is false.

1. Sum of the angles of a triangle is 180°. (P), T

2. $3+6=9$ T, P

3. $2-x=1$? [Not declarative sent.] → open statement [value of x is unknown]

4. Banglore is the capital of India. (F) P

5. Please come in. Not P

6. 3 is a prime no. P

7. Every rectangle is a square. (Not P)

8. What a beautiful evening. Not P

③ This ~~propos~~ has no truth value (T/F) provided it can be assigned the truth value T or F if the value of x is known. Hence it is not a statement or proposition. So these types of statements are called open statement.

* The propositions are represented by small letters like p, q, r, s, t, etc

Eg:- p : Rose is red

q : Rose is flower

The statement of ~~this~~ the form p & q are called simple statement.

Basic connectives : that turn simple stat to compound

① Conjunction logical AND ④ Implication

② Disjunction " OR ⑤ Bi-conditional

③ Negation NOT, NOR, NO

Eg:- Rose is red and rose is flower.

- 2 simple statements are connected by 'and', this statement is called a compound statement. Here 'and' is the connective. Ully there are other connectives like OR (disjunction), NOT (Negation), NOR, if then (implication), if and only if (bi-conditional).

① Conjunction : If 2 propositions p, q are connected by 'and', then it is called conjunction and it is denoted by the symbolic representn of conjunction, ie, $p \wedge q$.

② The compund proposition $p \wedge q$ is true when both p and q are true otherwise it is false.

Truth table:

p	q	$p \wedge q$
T	T	T
F	T	F
T	F	F
F	F	F

Eg:- p : $\sqrt{2}$ is an irrational no. (T)

q : 9 is a prime no. (F)

r : All triangles are equilateral (F)

s : 2+5=7 (T)

$p \wedge q$: $\sqrt{2}$ is an irrational no. and 9 is a prime no. (F)

$s \wedge p$: 2+5=7 and $\sqrt{2}$ is an irrational no. (T)

$r \wedge s$: All Δs are equilateral and 2+5=7 (F)

$q \wedge r$: 9 is a prime no. and All Δs are equilateral. (F)

③ Disjunction : Also called as Logical OR

If 2 propositions p and q are connected by 'OR' then it is called disjunction and it is denoted by $p \vee q$.

The compund statement $p \vee q$ is true if atleast one of p or q is true.

It is false when both p and q are false.

Truth table

p	q	$p \vee q$
T	T	T
F	T	T
T	F	T
F	F	F

$p \vee q \rightarrow T$

$p \vee s \rightarrow T$

$q \vee r \rightarrow F$

$r \vee s \rightarrow T$

Conditional / Implication :

If 2 propositions p and q are connected by the words **if p then q**, it is called as conditional / implication and it is denoted by $p \to q$.

Here p is called anticident / hypothesis.

q is called consequent / conclusion.

Truth table

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

The compand statement $p \to q$ is false, if p is true and q is false and it is true in all other cases.

Biconditional / Biimplication / Equivalence :

If 2 propositions p, q are connected by the words if and only if, then it is called biconditional & it is denoted by the symbol $p \leftrightarrow q$ or $p \Leftrightarrow q$.

$p \leftrightarrow q$ is true if p & q both are the same otherwise it is false.

Truth table

p	q	$p \leftrightarrow q$
T	T	T
F	T	F
T	F	F
F	F	T

1) p : 2 is a prime no. (T) r : 6 is a perfect square. (F)
q : 3 is a prime no. (T) s : 9 is a multiple of 6. (F)

$p \to q = T$ $\quad r \to p = T$ $\quad p \leftrightarrow q = T$ $\quad p \leftrightarrow r = F$ $\quad q \leftrightarrow r = F$

$p \to r = F$ $\quad r \to s = T$ $\quad r \leftrightarrow s = T$ $\quad p \leftrightarrow s = F$ $\quad q \leftrightarrow s = F$

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2) construct the truth table for the following compound propositions

i) $p \wedge (\sim q)$ (ii) $(\sim p) \vee q$ (iii) $p \to (\sim q)$ (iv) $(\sim p) \underline{\vee} (\sim q)$ ↓ X-OR opp to biconditional

i) $p \wedge \sim q$

p	q	$\sim q$	$p \wedge \sim q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

(ii) $(\sim p) \vee q$

p	q	$\sim p$	$(\sim p) \vee q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

iii) $p \to \sim q$

p	q	~q	p→~q
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	T

iv) $(\sim p) \veebar (\sim q)$

p	q	~p	~q	(~p) ⊻ (~q)
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	F

3) Let p and q be primitive statements for which the conditional $p \to q$ is false. Determine the truth values of the following compound propositions.

i) $p \wedge q$ F

ii) $(\sim p) \vee q$ F

iii) $q \to p$ T

iv) $(\sim q) \to (\sim p)$ F

$p \to q$
T F = F

4) Let p, q and r be propositions having the truth values 0, 0 and 1 resp. Find the truth values of the following compound propositions.

p q r
F F T

i) $(p \vee q) \vee r$ (T)

ii) $(p \wedge q) \wedge r$ (F)

iii) $(p \wedge q) \to r$ T

iv) $p \to (q \wedge r)$ T

v) $p \wedge (r \to q)$ F

vi) $p \to (q \to (\sim r))$ T

5) Construct the truth tables for the following compound propositions.

i) $(p \vee q) \wedge r$ ii) $p \vee (q \wedge r)$ iii) $(p \wedge q) \to \sim r$ iv) $q \wedge (\sim r \to p)$

p	q	r	~r	(p∨q)	(p∨q)∧r	q∧r	p∨(q∧r)	(p∧q)→~r	q∧(~r→p)
T	T	T	F	T	T	T	T	F	T
T	T	F	T	T	F	F	T	T	T
T	F	T	F	T	T	F	T	T	F
T	F	F	T	T	F	F	T	T	F
F	T	T	F	T	T	T	T	F	T
F	T	F	T	T	F	F	F	T	F
F	F	T	F	F	F	F	F	T	F
F	F	F	T	F	F	F	F	T	F

Tautology: A compound proposition which is true for all possible truth values of its components is called a tautology or a logical truth or a universally valid statement.

Contradiction: A comp. proposition which is false for all possible truth values of its components is called contradiction or an absurdity.

Contingency: A comp. proposition that can be true or false depending upon the truth values of its components is called a contingency. In other words, a contingency is a comp. proposition which neither a tautology nor a contradiction.

Eg:-

P	~P	P ∨ ~P	P ∧ ~P
T	F	T	F
F	T	T	F

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1) Show that for any 2 propositions p and q (i) $(p \vee q) \vee (p \leftrightarrow q)$ is a tautology.

(ii) $(p \vee q) \wedge (p \leftrightarrow q)$ is a contradiction.

(iii) $(p \vee q) \wedge (p \rightarrow q)$ is a contingency.

p	q	$p \vee q$	$p \leftrightarrow q$	$p \rightarrow q$	$(p \vee q) \vee (p \leftrightarrow q)$	$(p \vee q) \wedge (p \leftrightarrow q)$	$(p \vee q) \wedge (p \rightarrow q)$
T	T	F	T	T	T	F	F
T	F	T	F	F	T	F	F
F	T	T	F	T	T	F	T
F	F	F	T	T	T	F	F

---

2) Show that for any propositions p and q the compound proposition.

$p \rightarrow (p \vee q)$ is a tautology and the compound prop. $p \wedge (\sim p \wedge q)$ is contradiction.

p	q	$p \vee q$	$p \rightarrow (p \vee q)$	~p	$\sim p \wedge q$	$p \wedge \sim p \wedge q$
T	T	T	T	F	F	F
T	F	T	T	F	F	F
F	T	T	T	T	T	F
F	F	F	T	T	F	F

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3) S.T. the truth values of the following compound propositions are independent of their truth values. ~~of their~~

i) $\{p \wedge (p \to q)\} \to q$

ii) $(p \to q) \leftrightarrow (\sim p \vee q)$

p	q	$\sim p$	$p \to q$	$p \wedge (p \to q)$	$\{p \wedge (p \to q)\} \to q$	$\sim p \vee q$	$(p \to q) \leftrightarrow (\sim p \vee q)$
T	T	F	T	T	T	T	T
T	F	F	F	F	T	F	T
F	F	T	T	F	T	T	T
F	F	T	T	F	T	T	T

4) P.T. for any propositions p and q the compound proposition $[(\sim q) \wedge (p \to q)] \to \sim p$ is a tautology

5) P.T. for any proposition p, q, and r, $[(p \to q) \wedge (q \to r)] \to (p \to r)$ is a tau

and $\{p \to (q \to r)\} \to \{(p \to q) \to (p \to r)\}$ is a

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Logical equivalence:

Two proposition u and v are said to be logically equivalent whenever u and v have the same truth value or equivalently the biconditional $u \leftrightarrow v$ is a tautology.

$u \leftrightarrow v$ or $u \Leftrightarrow v$. When the propos^n u and v are not logically equivalent, then it is represented as $u \not\Leftrightarrow v$. Logically equivalent propositions are treated as identical propositions.

*1) Let x be a +ve integer, consider the following propositions

p: x is an odd integer

q: x is not divisible by 2.; Prove p and q are logically equivalent.

p and q have the same truth values so p and q are logically equivalent.

$p \to q \Leftrightarrow \sim p \vee q$

3) P.T. $\{(p \to q) \to r\} \Leftrightarrow \{(\sim p \vee q) \to r\}$

* $p \to q \leftrightarrow \sim p \vee q$

so $\{(\sim p \vee q) \to r\} \leftrightarrow \{(\sim p \vee q) \to r\}$

p	q	$p \to q$	$\sim p$	$\sim p \vee q$
T	T	T	F	T
F	T	T	T	T
T	F	F	F	F
F	F	T	T	T

2) P.1. for any propositions p and q, $p \veebar q$ and $(p \vee q) \wedge (\sim p \vee \sim q)$ are logically equivalent.

p	q	$p \veebar q$	$p \vee q$	$\sim p$	$\sim q$	$(\sim p \vee \sim q)$	$(p \vee q) \wedge (\sim p \vee \sim q)$
T	T	F	T	F	F	F	F
F	T	T	T	T	F	T	T
T	F	T	T	F	T	T	T
F	F	F	F	T	T	T	F

$\therefore p \vee q \leftrightarrow \{(p \vee q) \wedge (\sim p \vee \sim q)\}$

3) P.1. for any prop.s. p, q, r $\Rightarrow [p \to (q \wedge r)] \Leftrightarrow [(p \to q) \wedge (p \to r)]$

ii) $[(p \vee q) \to r] \Leftrightarrow [(p \to r) \wedge (q \to r)]$

iii) $[(p \vee q) \to r] \leftrightarrow [\sim r \to \sim (p \vee q)]$

i)

p	q	r	$q \wedge r$	$p \to (q \wedge r)$	$p \to q$	$p \to r$	$(p \to q) \wedge (p \to r)$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	F	F	T	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

$\therefore$ (i) proved.

ii)

p	q	r	$p \vee q$	$(p \vee q) \to r$	$p \to r$	$q \to r$	$(p \to r) \wedge (q \to r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

$\therefore$ (ii) proved

Laws of logic

① Law of double negation

$\sim\sim p \Leftrightarrow p$

② Idempotent laws

a) $(p \vee p) \Leftrightarrow p$ (b) $(p \wedge p) \Leftrightarrow p$

③ Identity laws

a) $(p \vee F_0) \Leftrightarrow p$ (F_0 = F) (b) $(p \wedge T_0) \Leftrightarrow p$ (T_0 = T)

④ Inverse Laws

a) $(p \vee \sim p) \Leftrightarrow T$ b) $(p \wedge \sim p) \Leftrightarrow F$

⑤ Domination laws

a) $(p \vee T) \Leftrightarrow T$ b) $(p \wedge F) \Leftrightarrow F$

⑥ Commutative laws

a) $(p \vee q) \Leftrightarrow (q \vee p)$ b) $(p \wedge q) \Leftrightarrow (q \wedge p)$

⑦ Absorption laws

a) $[p \vee (p \wedge q)] \Leftrightarrow p$ b) $[p \wedge (p \vee q)] \Leftrightarrow p$

⑧ Demorgan's laws

a) $\sim(p \vee q) \Leftrightarrow (\sim p \wedge \sim q)$ b) $\sim(p \wedge q) \Leftrightarrow (\sim p \vee \sim q)$

⑨ Associative laws

a) $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$ b) $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$

⑩ Distributive laws

a) $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$

b) $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

Law of negation for conditional

p	q	$p \to q$	$\sim(p \to q)$	$\sim q$	$p \wedge \sim q$
0	0	1	0	1	0
0	1	1	0	0	0
1	0	0	1	1	1
1	1	1	0	0	0

$\therefore \sim(p \to q) \Leftrightarrow p \wedge \sim q$

$\sim(p \vee q) \Leftrightarrow (\sim p \wedge \sim q)$

$\sim(p \wedge q) \Leftrightarrow (\sim p \vee \sim q)$

$\sim p \Leftrightarrow p$

Eg:- 1) Let x be a specified number, write down the negation of the following conditional:

If x is an integer, then x is a rational number.

soln

p: x is an integer

q: x is a rational number.

Given $p \to q$

Negation $\sim(p \to q) \Leftrightarrow p \wedge \sim q$

Ans:- x is an integer and x is not a rational number.

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2) Let x be a specified no. write down the negation of the following proposition

'If x is not a real no, then it is not a rational number and not an irrational no.'.

p: x is ~~not~~ a real no.

q: x is ~~not~~ a rational number

r: x is ~~not~~ an irrational number

$\sim p \to (\sim q \wedge \sim r)$

Negation:- $\sim[\sim p \to (\sim q \wedge \sim r)]$

$= \sim p \wedge [\sim(\sim q \wedge \sim r)]$

$= \sim p \wedge (q \vee r)$

→ x is not a real no and x is a rational no. or x is an irrational number.

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3) simplify the following compound proposition using Laws of Logic

(1) $(p \vee q) \wedge \sim\{(\sim p \vee q)\}$

~~(#)~~ $(p \vee q) \wedge (p \wedge \sim q)$. Demorgan's Law

$\{(p \vee q) \wedge p\} \wedge \sim q$ → Absorption Law (Associative)

$\Leftrightarrow \{p \wedge (p \vee q)\} \wedge \sim q$ → commutative Law

$\Leftrightarrow p \wedge (\sim q)$ .... Absorption Law

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2) $\sim[\sim\{(p \vee q) \wedge r\} \vee \sim q]$

$\Leftrightarrow \sim(\sim\{(p \vee q) \wedge r\}) \wedge \sim(\sim q)$ DeMorgn's Law

$\Rightarrow [(p \vee q) \wedge r] \wedge q$ Law of Double Negation (Assoc)

$\Rightarrow (p \vee q) \wedge (r \wedge q)$ Associative Law

$\Rightarrow (p \vee q) \wedge (q \wedge r)$ Commutative Law

$\Rightarrow [(p \vee q) \wedge q] \wedge r$ Associative Law of $\wedge$

$\Rightarrow q \wedge r$ Absorption law, Commutative law

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1) $(p \vee q) \wedge \sim\{(\sim p \wedge q) \Leftrightarrow p$

given:- $(p \vee q) \wedge \sim(\sim p \wedge q)$

$\Leftrightarrow (p \vee q) \wedge (\sim\sim p \vee \sim q)$ → by using Demorgan's law

$\Leftrightarrow (p \vee q) \wedge (p \vee \sim q)$ → Law of double negation

$\Leftrightarrow p \vee (q \wedge \sim q)$ → Distributive Law

$\Leftrightarrow p \vee F$ → Inverse Law

$\Leftrightarrow p$

2) $(p \vee q) \wedge \sim\{(\sim p \vee q)$

$\Leftrightarrow (p \vee q) \wedge (\sim\sim p \wedge \sim q)$ ... Demorgan's law

$\Leftrightarrow (p \vee q) \wedge (p \wedge \sim q)$ ..... Law of double negation

$\Leftrightarrow \{(p \vee q) \wedge p\} \wedge \sim q$ ..... Associative law

$\Leftrightarrow \{p \wedge (p \vee q)\} \wedge \sim q$ ..... Commutative Law

$\Leftrightarrow p \wedge \sim q$ ..... Absorption Law

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3) $p \vee [p \wedge (p \vee q)] \Leftrightarrow p$

$p \vee [p \wedge (p \vee q)]$

$\Leftrightarrow p \vee p$ ..... Absorption Law

$\Leftrightarrow p$ ..... Idempotent law

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4) $[p \vee q \vee (\sim p \wedge \sim q \wedge r)] \Leftrightarrow p \vee q \vee r$

We have $\sim p \wedge \sim q \wedge r$

$\Leftrightarrow (\sim p \wedge \sim q) \wedge r$ ..... Associative Law

$\Leftrightarrow \sim(p \vee q) \wedge r$ ..... By demorgan's law

Now $[p \vee q \vee (\sim p \wedge \sim q \wedge r)]$

$\Leftrightarrow [p \vee q \vee \{\sim(p \vee q) \wedge r)]$

$\Leftrightarrow [(p \vee q) \vee \sim(p \vee q)] \wedge [(p \vee q) \vee r]$ ..... Distributive law

$\Leftrightarrow T \wedge [(p \vee q) \vee r]$ ..... Inverse Law

$\Leftrightarrow [(p \vee q) \vee r] \wedge T$ ..... Commutative Law

$\Leftrightarrow p \vee q \vee r$ ..... Identity Law

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5) $[(\sim p \vee \sim q) \rightarrow (p \wedge q \wedge r)]$

we have $p \rightarrow q \Leftrightarrow \sim p \vee q$

$\Leftrightarrow \sim(\sim p \vee \sim q) \vee (p \wedge q \wedge r)$

$\Leftrightarrow (\sim\sim p \wedge \sim\sim q) \vee (p \wedge q \wedge r)$ ... Demorgan's Law

$\Leftrightarrow (p \wedge q) \vee (p \wedge q \wedge r)$ ... Law of double negation

$\Leftrightarrow p \wedge q$ ... From absorption law [$p \vee (p \wedge q) \Leftrightarrow p$]

6) $[(p \vee q) \wedge (p \vee \sim q)] \vee q \Leftrightarrow p \vee q$

$\Leftrightarrow [p \vee (q \wedge \sim q)] \vee q$ ..... Distributive Law

$\Leftrightarrow [p \vee F] \vee q$ ..... Inverse Law

$\Leftrightarrow p \vee q$ ..... Identity Law

7) $(p \rightarrow q) \wedge (\sim q \wedge (r \vee \sim q)) \Leftrightarrow \sim(q \vee p)$

$(p \rightarrow q) \wedge (\sim q \wedge (r \vee \sim q))$

$(p \rightarrow q) \wedge (\sim q \wedge (\sim q \vee r))$ ... Commutative law

$(p \rightarrow q) \wedge (\sim q)$ ..... Absorption Law

$(\sim p \vee q) \wedge (\sim q)$ ..... From $p \rightarrow q \Leftrightarrow \sim p \vee q$

$(\sim q) \wedge (\sim p \vee q)$ ..... Commutative Law

$(\sim q \wedge \sim p) \vee (\sim q \wedge q)$ ... Distributive law

$(\sim q \wedge \sim p) \vee F$ ..... Inverse Law

$\sim q \wedge \sim p$ ..... Identity Law

$\Leftrightarrow \sim(q \vee p)$ ..... Demorgan's Law

1) $[(p \vee q) \wedge (p \vee \sim q)] \vee q \Leftrightarrow p \vee q$

$\Leftrightarrow p \vee (q \wedge \sim q) \vee q$ ... Distributive law

$\Leftrightarrow (p \vee F) \vee q$ ..... Contradiction

$\Leftrightarrow p \vee q$ ..... Identity law

2) ~~$(p \rightarrow q) \wedge [\sim q \wedge (r \vee \sim q)] \Leftrightarrow \sim(q$~~

2) $(p \vee q) \wedge \sim\{(\sim p) \vee q\}$

$\Leftrightarrow (p \vee q) \wedge (\sim(\sim p) \wedge \sim q)$ .... De-morgan's law

$\Leftrightarrow (p \vee q) \wedge (p \wedge \sim q)$ .... Law of double negation

$\Leftrightarrow ((p \vee q) \wedge p) \wedge \sim q$ .... Associative Law

$\Leftrightarrow p \wedge \sim q$ .... Absorption law

3) $p \to (q \to r) \Leftrightarrow (p \wedge q) \to r$

$p \to (q \to r)$

$\Leftrightarrow p \to (\sim q \vee r)$ ---- Use $p \to q \Leftrightarrow \sim p \vee q$

$\Leftrightarrow \sim p \vee (\sim q \vee r)$

$\Leftrightarrow (\sim p \vee \sim q) \vee r$ .... Associative Law

$\Leftrightarrow \sim(p \wedge q) \vee r$ .... Demorgan's law

$\Leftrightarrow (p \wedge q) \to r$

4) $[\sim p \wedge (\sim q \wedge r)] \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$

$\Leftrightarrow [(\sim p \wedge \sim q) \wedge r] \vee [(q \wedge r) \vee (p \wedge r)]$ .... Associative law

$\Leftrightarrow [\sim(p \vee q) \wedge r] \vee [r \wedge (q \vee p)]$ .... Demorgan's Law & Distributive

$\Leftrightarrow [\sim(p \vee q) \wedge r] \vee [(q \vee p) \wedge r]$ .... Commutative law

$\Leftrightarrow [\sim(p \vee q) \wedge r] \vee [(p \vee q) \wedge r]$

$\Leftrightarrow r \wedge [(\sim(p \vee q)) \vee (p \vee q)]$ .... Distributive law

$\Leftrightarrow r \wedge T$ .... Tautology

$\Leftrightarrow r$ .... Identity law

5) $\sim[\{(p \vee q) \wedge r\} \to \sim q]$

$\Leftrightarrow \sim[\sim\{(p \vee q) \wedge r\} \vee \sim q]$

$\Leftrightarrow q \wedge r$ —— ② problem

6) $[(p \vee q) \wedge r] \vee \sim q \Leftrightarrow q \wedge r$

$p \to q \Leftrightarrow \sim p \vee q$

6) $[(p \vee q) \wedge r] \vee \neg q \Leftrightarrow q \wedge r$

$(p \wedge r) \vee (q \wedge r) \vee \neg q$ ..... Distributive law

$r \vee \neg q$

Applications of switching networks :

A switching network is made up of wires and switches connecting 2 terminals A and B.

A — P — B

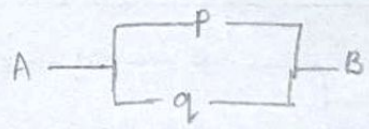
In a such a network, each switch is open (no current flows through it or closed (current flows through it). If a switch P is open, we assign a symbol 0 to it and if it is closed, then we assign a symbol 1 to it.

① Serial network → consisting of 2 switches p and q in which the current flows from the terminal A to the terminal B if both P and q are closed, then we assign a symbol 1 to it. The network is represented as $p \wedge q$.

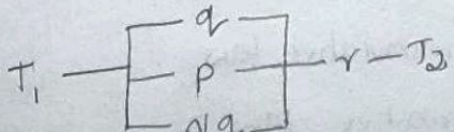
A — P — q — B

② Parallel network :

It represents the parallel network consisting of 2 switches p and q in which the current flows from the terminal A to B, if P or q is closed OR both p, q are closed are assigned by the value of 1. The network is represented as $p \vee q$.



* simplify the switching network given below



$\Rightarrow T_1 — r — T_2$

$(q \vee p \vee \neg q) \wedge r$

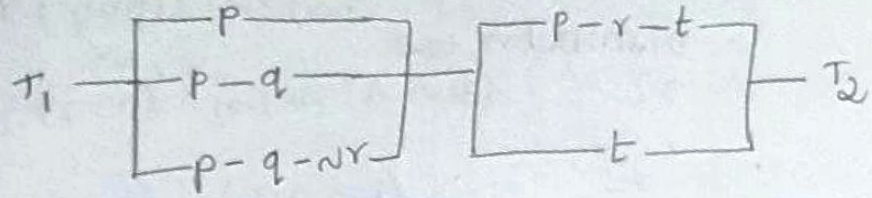
$\Leftrightarrow \{(q \vee \neg q) \vee p\} \wedge r$ .... Associative law

$\Leftrightarrow (T \vee p) \wedge r$ ..... Inverse law

$\Leftrightarrow T \wedge r$ ... Domination law

$\Leftrightarrow r$ ..... Identity law

2)



$\{p \vee (p \wedge q) \vee (p \wedge q \wedge \sim r)\} \wedge \{(p \wedge r \wedge t) \vee t\}$

$\{p \vee [p \wedge (q \wedge \sim r)]\} \wedge \{(p \wedge r \wedge t) \vee t\}$ ... Absorption law and associative law

$p \wedge \{(p \wedge r \wedge t) \vee t\}$ ..... Absorption law

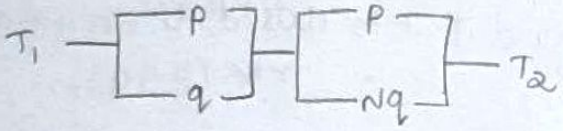
$p \wedge \{t \vee [t \wedge (p \wedge r)]\}$ ..... Commutative and associative law

$p \wedge t$ ..... Absorption law.

T_1 — p — t — T_2

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3)



$(p \vee q) \wedge (p \vee \sim q)$

$\Leftrightarrow$ $p \vee (q \wedge \sim q)$ ..... Distributive law

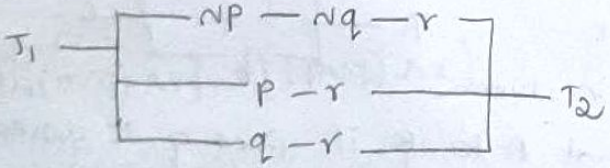
$\Leftrightarrow$ $p \vee F$ ..... Inverse Law

$\Leftrightarrow$ p ..... Identity law

T_1 — p — T_2

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4)



T_1 — r — T_2

$(\sim p \wedge \sim q \wedge r) \vee [(p \wedge r) \vee (q \wedge r)]$

$\Leftrightarrow$ $(\sim p \wedge \sim q \wedge r) \vee [r \wedge (p \vee q)]$ ..... Distributive law

$\Leftrightarrow$ $(\sim (p \vee q) \wedge r) \vee [r \wedge (p \vee q)]$ ..... De-morgan's Law

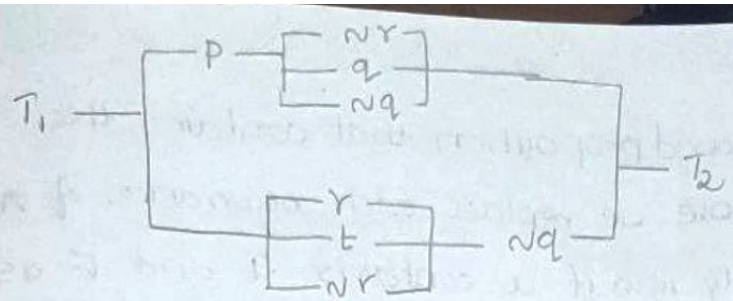
$\Leftrightarrow$ $[\sim (p \vee q) \wedge r] \vee [(p \vee q) \wedge r]$ ..... Commutative law

$\Leftrightarrow$ $r \wedge [\sim (p \vee q) \vee (p \vee q)]$ ..... Distributive law

$\Leftrightarrow$ $r \wedge T$ ..... Inverse law

$\Leftrightarrow$ r ..... Identity law

5)



$[p \wedge (\neg r \vee q \vee \neg q)] \vee [(r \vee t \vee \neg r) \wedge \neg q]$

$\Leftrightarrow \{p \wedge [\neg r \vee (q \vee \neg q)]\} \vee \{[t \vee (r \vee \neg r)] \wedge \neg q\}$ ... Associative law

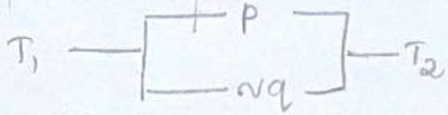
$\Leftrightarrow \{p \wedge [\neg r \vee T]\} \vee \{[t \vee T] \wedge \neg q\}$ ... Inverse law

$\Leftrightarrow \{p \wedge T\} \vee \{T \wedge \neg q\}$ ... Domination law

~~$\Leftrightarrow \{T \wedge p\} \vee \{T \wedge \neg q\}$ ... Commutative law~~

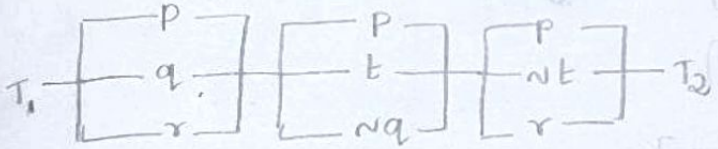
~~$\Leftrightarrow T \vee (p$~~

$\Leftrightarrow p \vee \neg q$ ... Identity law



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6)



$(p \vee q \vee r) \wedge (p \vee t \vee \neg q) \wedge (p \vee \neg t \vee r)$

$(p \vee q \vee r) \wedge (p \vee \neg t \vee \neg t$

Duality:

Suppose u is a compand proposition that contains the connectives $\wedge$ and $\vee$, suppose we replace each occurrances of $\wedge$ and $\vee$ by $\vee$ and $\wedge$ respectively. Also if u contains T and F as components, suppose we replace each occurrances of T and F by F and T respectively, then the resulting compand proposition is called dual of u and denoted by u^d.

Eg:- 1. $u = \{(p \vee q) \wedge (s \wedge t) \wedge F\}$ 2. $u = p \wedge (q \vee \sim r) \vee (s \wedge T)$

$u^d = \{(p \wedge q) \vee (s \vee t) \vee F\}$ $u^d = p \vee (q \wedge \sim r) \wedge (s \vee F)$

For the following 2 results are...

1. the dual of u is logically equivalent to u $*(u^d)^d \Leftrightarrow u$
2. For any proporitions u and v, if $u \Leftrightarrow v$ then $u^d \Leftrightarrow v^d$ this is known as the principle of duality.

Find the dual of

1) $\sim(p \vee q) \wedge [p \vee \sim(q \wedge \sim s)]$

Dual: $\sim(p \wedge q) \vee [p \wedge \sim(q \vee \sim s)]$

2) $[(p \vee T) \wedge (q \vee F)] \vee (r \wedge s) \wedge T]$

Dual: $[(p \wedge F) \vee (q \wedge T)] \wedge (r \vee s) \vee F]$

3) $[(p \wedge q) \vee [(\sim(p \vee q) \wedge (\sim r \wedge s)] \vee (r \vee s)]$

Dual: $[(p \vee q) \wedge (\sim p \wedge q) \vee (\sim r \vee s)] \wedge (r \wedge s)]$

4) $p \to q$
$\Leftrightarrow (\sim p \vee q)^d$
$\Leftrightarrow (\sim p \wedge q)$

5) $(p \to q) \to r$
$\Leftrightarrow (\sim(p \to q) \vee r)^d$
$\Leftrightarrow (\sim(\sim p \wedge q) \vee r)^d$
$\Leftrightarrow (p \vee \sim q) \wedge r$

6) $p \to (q \to r)$
$\Leftrightarrow [\sim p \vee (q \to r)]^d$
$\Leftrightarrow [\sim p \vee (\sim q \vee r)]^d$
$\Leftrightarrow (\sim p \wedge (\sim q \wedge r))$

7) $[(\sim p \vee q) \wedge (p \wedge (p \wedge q))] \Leftrightarrow p \wedge q$

Dual: $[(\sim p \wedge q) \vee (p \vee (p \vee q))] \Leftrightarrow p \vee q$

quesh-

$\Leftrightarrow [(\sim p \wedge q) \vee (p \vee p) \vee q]$

$\Leftrightarrow [(\sim p \wedge q) \vee (p \vee q)]$

$\Leftrightarrow [\sim p \wedge (p \vee q) \vee q \wedge (p \wedge q)]$

$[(\sim p \vee q) \wedge (p \wedge (p \wedge q)]$

$[(\sim p \vee q) \wedge (p \wedge p) \wedge q]$

$(\sim p \vee q) \wedge (p \wedge q)$

$\Leftrightarrow (F \wedge q) \vee (q \wedge p)$

$\Leftrightarrow F \vee (q \wedge p)$

$\Leftrightarrow F \vee (p \wedge q)$

$\Leftrightarrow p \wedge q$

8) $[\sim (p \wedge q) \rightarrow \sim p \vee (\sim p \vee q)] \Leftrightarrow (\sim p \vee q)$

$u \rightarrow v \Leftrightarrow \sim u \vee v$

Dual: $\sim[\sim(p \vee q)] \vee (\sim p \vee (\sim p \vee q))$

$(p \vee q) \vee (\sim p \vee (\sim p \vee q))$

Dual: $(p \vee q) \wedge (\sim p \wedge (\sim p \wedge q)) \Leftrightarrow (\sim p \wedge q)$

Now: $(p \vee q) \vee (\sim p \vee (\sim p \vee q))$

$\Leftrightarrow (p \vee q) \vee ((\sim p \vee \sim p) \vee q)$

$\Leftrightarrow (p \wedge q) \vee (\sim p \vee q)$

$\Leftrightarrow ((p \wedge q) \vee q) \vee \sim p$ — Associative

$\Leftrightarrow q \vee \sim p$ — Absorption

$\Leftrightarrow \sim p \vee q$ — Commutative

Dual of RHS = $\sim p \wedge q$ (verified)

The connectives NAND and NOR:

For any 2 proposition p and q, Demorgan's law states that

$\sim(p \wedge q) \Leftrightarrow (\sim p \vee \sim q)$

$\sim(p \vee q) \Leftrightarrow (\sim p \wedge \sim q)$

The comp. proposition negation $\sim(p \wedge q)$ is read as $\sim(p \wedge q)$ denoted by $p \uparrow q$

$\uparrow \rightarrow$ NAND connective.

The comp proposition $\sim(p \vee q)$ denoted by $p \downarrow q$.

$\downarrow \rightarrow$ NOR connective.

p	q	$p \uparrow q$	$p \downarrow q$
0	0	1	1
0	1	1	0
1	0	1	0
1	1	0	0

$p \uparrow q$ and $p \downarrow q$ are duals of each other

For any propositions p, q, prove:

i) $\sim(p \downarrow q) \Leftrightarrow (\sim p \uparrow \sim q)$

$\sim(\sim(p \vee q))$

$\Leftrightarrow \sim(\sim p \wedge \sim q)$

$\Leftrightarrow \sim p \uparrow \sim q$?

ii) $\sim(p \uparrow q) \Leftrightarrow (\sim p \downarrow \sim q)$

$\Leftrightarrow \sim\{\sim(p \wedge q)\}$

$\Leftrightarrow \sim(\sim p \vee \sim q)$

$\Leftrightarrow \sim p \downarrow \sim q$

iii) $p \uparrow (q \uparrow r) \Leftrightarrow \sim p \vee (q \wedge r)$

$\Leftrightarrow \sim[p \wedge (q \uparrow r)]$

$\Leftrightarrow \sim[p \wedge \sim(q \wedge r)] \Leftrightarrow \sim p \vee (q \wedge r)$

iv) $(p \uparrow q) \uparrow r \Leftrightarrow (p \wedge q) \vee \sim r$

$\Leftrightarrow \sim[(p \uparrow q) \wedge r]$

$\Leftrightarrow \sim[\sim(p \wedge q) \wedge r] \Leftrightarrow (p \wedge q) \vee \sim r$

v) $p \downarrow (q \downarrow r) \Leftrightarrow \sim p \wedge (q \vee r)$

$\Leftrightarrow \sim[p \vee (q \downarrow r)]$

$\Leftrightarrow \sim[p \vee \sim(q \vee r)] \Leftrightarrow \sim p \wedge (q \vee r)$

vi) $(p \downarrow q) \downarrow r \Leftrightarrow (p \vee q) \wedge \sim r$

$\Leftrightarrow \sim[(p \downarrow q) \vee r]$

$\Leftrightarrow \sim[\sim(p \vee q) \vee r] \Leftrightarrow (p \vee q) \wedge \sim r$